Circles Base Group Mint Router Audit

Gnosis Ltd - Report by Côme du Crest

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Circles Base Group Mint Router Audit

This document presents the findings of a smart contract audit conducted by Côme du Crest for Gnosis Ltd.

Scope

The scope includes contracts modified in PR #29 of aboutcircles/circles-groups the last commit of the PR reviewed being 0x9d38f5a.

The repository has already been audited as of commit 0x1c32af9. The diff can be seen here

Context

When group tokens attempt to be minted through Hub.operateFlowMatrix(), the function _groupMint() will be called to mint the group tokens to the group itself in order to send them later following the matrix flows. However, the _groupMint() function uses safeBatchTransferFrom(_sender, ...) which requires the sender to have approved the msg.sender for all tokens:

```
    function safeBatchTransferFrom(
        address _from,
        address _to,
        uint256[] memory _ids,
        uint256[] memory _values,
        bytes memory _data
    ) public virtual {
        address sender = _msgSender();
        if (_from != sender && !isApprovedForAll(_from, sender)) {
            revert ERC1155MissingApprovalForAll(sender, _from);
        }
        _safeBatchTransferFrom(_from, _to, _ids, _values, _data);
    }
```

See Hub.sol.

This prevents _groupMint() to be used in a transfer such as A -> B -> Group -> C if B did not set approval to A (or msg.sender).

To circumvent this, a BaseGroupMintRouter contract has been implemented that will allow to trustlessly set approval to any human CRC tokens. Additionally, if the CRC is trusted by the underlying group the router will also trust it. That is isApprovedForAll(Router, CRC) and isTrusted(Router, CRC) can be set to true for any CRC. This will enable transfer such as A -> B ->

`Router -> Group -> C` since `Router` can be set to approve `A` (or any other human `msg.sender`).

The `BaseGroupMintRouter` is an organization (as defined per the `Hub`) and is not meant to hold CRC. As such there is no problem for the router to approve or trust any address. This is just a convenience to get around the Hub's restriction on group mints.

Status

No issue has been found.